MIL-T-21147(SHIPS)
AMENDMENT - 2
3 May 1960
SUPERSEDING
Amendment - 1
1 March 1960

MILITARY SPECIFICATION

TERMINAL, TELEGRAPH, AN/UGC-1( )

This amendment forms a part of Military Specification MIL-T-21147(SHIPS), 12 November 1957.

Page 1: Add as paragraphs 3.1.2 and 3.1.3:

"3.1.2 Printed circuitry shall be used as extensively as possible. Particular attention shall be paid to the printed circuit requirements of Specification MIL-E-16400.

"3.1.3 Modular construction shall be used throughout the equipment in accordance with the appendix of this specification."

Page 3, paragraph 3.2.10: Delete and substitute:

"3.2.10 Radio interference and undesired radiation. - The equipment shall meet the radio interference and undesired radiation requirements of Specification MIL-I-16910, except that any spurious radiated signals in the 14 kc. to 1,000 mc. range shall not exceed one microvolt per meter when measured at a distance of 3 feet in any and all directions from the equipment; and conducted interference shall not exceed one microvolt per kilocycle bandwidth as measured from line to ground."

Page 3, paragraph 3.4.1.5: Add the following sentences "Isolations of the input terminals shall not be obtained by mechanical relays. Semiconductor circuitry shall be utilized."

Page 4, paragraph 3.4.2.2.1: Delete and substitute:

"3.4.2.2.1 The transmitter output circuit shall be designed so that it may be keyed by 130 volts from an external source. The internal impedance shall not exceed 800 ohms."

Page 4, paragraph 3.4.2.2.1.1: Delete and substitute:

"3.4.2.2.1.1 The output circuit shall be ungrounded, and designed so that either side of the circuit may be grounded externally."

Page 4, paragraph 3.4.2.2.2, line 1: Delete "plus or minus 1 volt" and substitute "plus or minus 5 volts".

Page 4, paragraph 3.4.4.1: Delete and substitute:

"3.4.4.1 The receiving equipment start-stop output shall not be keyed by a mechanical relay. Semiconductor circuitry shall be used to key the output signal lines, and the output terminals shall be electrically isolated from chassis and cabinet. Line battery shall be supplied from an external source."

Page 5, paragraph 3.4.5.1.1.1, column 2, line 3: Delete "732 p.p.s." and substitute "733.69 p.p.s."

Page 6, paragraph 3.4.6.2: Add as items (c), (d), and (e):

"(c) Provisions for automatically detecting misframing at the receive terminal shall be incorporated.
(d) Audio and visual out of frame alarms shall be incorporated.
(e) Automatic lock up in a steady marking condition of all output loops during a misframe condition shall be incorporated."

FSC 5805

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Page 8, paragraph 4.2: Delete and substitute:

"4.2 Preproduction inspection.- Preproduction inspection shall be performed by the contractor and shall be in accordance with Specification MIL-E-16400 with the following exceptions:

(a) Test procedures.- A detailed list of proposed tests, test procedures, and test equipment shall be submitted to the bureau or agency concerned for approval at least 30 days prior to beginning tests.
(b) Parts inspection.- Prior to preproduction testing, the Government inspector shall examine all parts and semiconductors in the preproduction model to determine compliance with the parts requirements of Specification MIL-E-16400. If the model is found to contain any nonpreferred parts, which do not have the approval of the bureau or agency concerned, the inspector shall not release the model for testing until such corrections are necessary have been made.
(c) Accelerated life tests.- In lieu of the accelerated life test, a 15 day operating test shall be run at room temperatures. Operation shall be considered satisfactory if the error rate in each channel is less than 1 in 50,000 words.
(d) Preproduction model acceptance.- Acceptance of the preproduction model shall be contingent upon approval by the bureau or agency concerned of the preproduction inspection report, submitted in accordance with Specification MIL-E-16400."

Page 8, paragraph 4.3.1.1, lines 1 and 2: Delete "at random by the Government inspector".

Page 8: Add as paragraph 4.4:

"4.4 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

Preparing activity:
Navy - Bureau of Ships
(Project 5805-0001Sh)

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Add appendix:

"APPENDIX

10. SCOPE

10.1 This appendix establishes certain design features of unitized equipment which are necessary to provide a planned maintenance program for these equipments.

20. APPLICABLE DOCUMENTS

20.1 The following specifications, standards and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-C-21097 - Connectors, Plug and Receptacle, Painted Wiring Board.

STANDARDS

MILITARY
MIL-STD-439 - Electronic Circuits.
MIL-STD-706 - Power Supply Voltages for Electronic Equipment.
MS16108 - Jacks, Tip, Low Voltage.

PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS 900,152 - Manufacturers Designating Symbols.

BUREAU OF AERONAUTICS
NAVAER 16-1 519 - Handbook Preferred Circuits.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. DEFINITIONS

30.1 Assembly.- An assembly is a commonly-mounted group of parts combined to perform a specific electronic function, and capable of being easily removed and replaced as an integral item during routine equipment maintenance (also called "Subassembly"). Note: Certain items falling within the definition of assembly are considered for convenience to be parts because of similarities to simple parts in construction and electrical characteristics. Examples are IF transformers and line filters.

30.2 Modular assembly.- A modular assembly is an assembly whose outline dimensions are integral multiples of a module.

30.3 Module.- A module is a basic linear dimension.

30.4 Unitized equipment.- Unitized equipment is a type of equipment in which the electronic circuitry is partially or wholly contained in assemblies.

40. GENERAL REQUIREMENTS

40.1 Design objectives.- During equipment design, consideration shall be given to the following factors and their possible effect on the maintainability of the installed equipment:

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(a) Reduction of the number of different types of assembly construction within an equipment.
(b) Selection of maintenance design features compatible with the overall equipment complexity and cost.
(c) Variations in the maintenance capabilities of various ship types on which the equipment is intended to be installed.

40.2 Test points.- One or more test points shall be provided for each assembly regardless of type, so located in the circuit that a single measurement at that point will indicate assembly performance or failure. The test points shall be located on the top (that is, accessible) surface of the assembly, or shall be brought out through one or more pins of the connector (see 4.8), or both. One of the objectives in design shall be to limit the types of necessary test equipment to volt-ohmmeters and oscilloscopes. The measurement facility may be contained within the equipment, in which case external test points and external test equipment are not required. Test point jacks shall conform to Standard MS16108.

40.3 Technical manuals.- The technical manuals shall contain a simple, direct, unambiguous fault location procedure, which shall detail step-by-step the sequence of measurements to be made to locate a failed assembly. In addition, for each repairable assembly, the technical manual shall include a step-by-step procedure for locating failed parts within the assembly, using any combination of the following which will result in the simplest procedure:

(a) Resistance measurements.
(b) Voltage measurements, with power supplied to the assembly (see 4.4).
(c) Signal measurements, specifying the particular input conditions (see 4.4).

40.4 Test facilities.- The selection of maintenance procedures for assembly repairs described in the technical manual shall be based on the availability of the following test facilities:

(a) Volt ohmmeter.
(b) Oscilloscope.
(c) AF sine-wave oscillator.
(d) RF signal gen.
(e) Pulse or square wave gen.
(f) Teletypewriter distortion analyzer.
(g) Teletypewriter signal generator.
(h) A test chassis containing the three preferred connectors wired to internal power supplies and input-output connectors as shown on figure 1.

40.5 Repairability.- Assemblies shall be repairable on board ship without special tools. Unrepairable assemblies, and those requiring special facilities for repair, shall be used only when approved by the bureau or agency concerned.

40.6 Assembly size.- Modular assemblies shall be based on a module of 1.5 inches. Line-to-line dimensions of all assemblies shall be selected in accordance with table I.

Table I - Assembly sizes.

Height	Width	Length
4.5	1.5	4.5 preferred
4.5	3.0	4.5 preferred
4.5	4.5	4.5 preferred
4.5	6.0	4.5 preferred
3.0	0.75	3.0 preferred
3.0	1.5	3.0 preferred
4.5	1.5	6.0 acceptable
4.5	3.0	6.0 acceptable

Note 1: Width and length dimensions refer to the mounting surface of the assembly.
Note 2: All dimensions are in inches, and are line-to-line dimensions. Maximum values of the actual widths and lengths shall be 0.062 inch less than the line-to-line dimensions.

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40.7 Connectors. - Except for high voltage or high frequency leads, all electrical connections shall be made to the assembly through a single plug-in connector selected in accordance with table II, and located on the mounting surface of the assembly.

Table II - Connectors.

Cannon DA 15 P or equivalent	Preferred
Winchester MRE 14 P-G or equivalent	Panel-mounted units only
Type POLLd of MIL-C-21097	Preferred

40.8 Connector pin designations. - Electrical connections within the assembly shall be made to the connector in accordance with table III.

Table III - Connector pin designations.

Cannon DA 15 P or equivalent	Winchester W11P or equivalent	Winchester MRE 14 P-G or equivalent
1. Input #1	A. Input #1	A. Input #1
2. *Gnd or Input #3	B. Input #2	B. *Gnd or Input #3
3. +150 V	C. *Input #3	C. Input #2
4. 6.3 V (ac)	D. Gnd	D. Gnd
5. 6.3 V (ac)	E. +12 V	E. Gnd
6. Gnd	F. +25 V	F. +150 V
7. -150 V	H. -25 V	H. 6.3 V heater #1
8. Output #1	J. -12 V	J. +250 V, +300 V
9. Input #2	K. *Output #3	K. 6.3 V heater #2
10. *Gnd or Input #4	L. Output #2	L. 6.3 V heater #1
11. +300 V, +250 V	M. Output #1	M. -150 V
12. 6.3 V (ac)		N. 6.3 heater #2
13. 6.3 V (ac)		P. Output #1
14. Gnd		R. *Output #2
15. *Output #2		

Note. - Pins designated (*) may be used for test point connections.

40.9 Thermal design. - Metallic conduction paths shall be provided from major heat-producing parts to the assembly case, and from the case to the equipment chassis or supporting structure. Where additional cooling is necessary, assemblies shall be designed for air flow using the mounting surface for inlet and the top surface for exhaust.

40.10 Mounting methods. - Assemblies containing controls required during equipment operation shall be mounted by means of a front panel and captive knurl head screws, with the mounting surface parallel to the front panel. Other assemblies shall be mounted to permit easy removal and replacement without special tools and without removing other assemblies. A top surface strap and guide-pin arrangement is preferred.

40.11 Marking. - Assembly marking shall consist at least of the manufacturer's name or code, and the assembly functional description. Manufacturer's code designations shall be in accordance with Publication NAVSHIPS 900,152. All marking shall be located on the top surface of the assembly. Test point jacks shall be designated by an appropriate waveform symbol or voltage. The preferred method of marking shall be by stamping or engraving; the use of ink covered with clear lacquer is also acceptable.

40.12 Maintenance controls. - All controls required only during equipment maintenance shall be located on the top surface of the assembly.

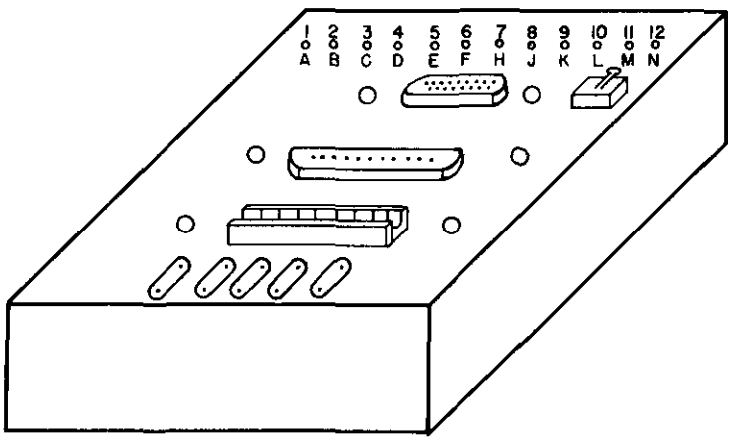
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40.13 Drawings. - Manufacturing drawings required by the contract or order shall include for each assembly the overall performance characteristics, the tolerance limits for the function, a complete parts list, and a schematic diagram.

40.14 Reliability prediction. - Using the methods described in Vitro Laboratory Report Nos. 80 and 98, or in RCA Report TR-1100, the MTBF (mean-time-between-failures) shall be computed for each assembly and submitted to the bureau or agency concerned for review and approval.

40.15 Circuit design. - Where applicable, circuits shall be selected from Standard MIL-STD-439 or Publication NAVAER 16-1 519. Preferred values of power supply voltages shall be 6, 12, 25, 150, 250, and 300 volts. Other values specified in Standard MIL-STD-706 are also acceptable."

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NOTES:

ooooo DENOTES CONNECTION JACKS TO EACH TERMINAL OF ALL CONNECTORS (EXCEPT FOR INPUT & OUTPUT).

O DENOTES INPUT & OUTPUT JACKS FOR EACH CONNECTOR.

DENOTES 250 VOLT/300 VOLT SWITCH, 6,3,25,150,250/300 VOLT POWER SUPPLIES, CONTAINED WITHIN THE TEST CHASSIS, ARE INTERNALLY WIRED TO THE SPECIFIED CONNECTOR PINS (SEE TABLE III) THROUGH JUMPERS DENOTED

FIGURE 1: TEST CHASSIS

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